

Chapter : More about solutions of systems of linear equations

Section 7.1: The space of solutions of a homogeneous system

We know that the set L of solutions of a *homogeneous* system of linear equations in n variables is a subspace of $\mathbb{R}^n$, see slide 21 of Chapter 3. We will see in this part of the module how we can determine a (finite) basis for L . The number of basis elements, that is, the dimension of L , will be seen to equal the number of parameters as determined by Gauss elimination. In particular, *the number of parameters does not depend on how Gauss elimination is performed.*

1 Solutions of homogeneous systems of linear equations: finding a basis

Let $L \subseteq \mathbb{R}^n$ be the set of solutions of a homogeneous system of linear equations in n variables.

We know that L is a *subspace* of $\mathbb{R}^n$ (slide 21 of Chapter 3). Hence L is a finitely generated vector space with $\dim(L) \leq n$ (see slide 28 of Chapter 6).

But we can do better than that!

Aim: Find a basis for L , and interpret the number $\dim(L)$.

2 The dimension of the space of solutions of a homogeneous systems of linear equations

For a homogeneous system of linear equations in n variables $x_1, x_2, \dots, x_n$, let L denote its space of solution. Then:

- The vector space L is finitely generated.
- $\dim(L) \leq n$
- $\dim(L)$ is precisely the number k of parameters as determined by Gauss elimination.

We know the first two statements already, see previous slide.

To prove the last, we will produce an explicit basis for L , and this basis will have k elements. See next slide for the algorithm!

3 Finding a basis for the space of solutions of a homogeneous systems of linear equations: Algorithm

Given a homogeneous system of linear equations in n variables $x_1, x_2, \dots, x_n$, we find a basis for the set L of solutions as follows:

- Transform system into row echelon form, let k = number of parameters ($0 \leq k \leq n$).
- If $k = 0$: System has unique solution, so $L = \{\underline{0}\}$. Basis for L : the empty set.
- If $k > 0$: define $b_j \in \mathbb{R}^n$, for $j = 1, 2, \dots, k$:

Set the j th parameter to 1, and all other parameters to 0.
Calculate the non-parameters. These specific choices of $x_1, \dots, x_n$ are the entries of the vector b_j .

The vectors $b_1, b_2, \dots, b_k$ form a basis for L .

4 Space of solutions of a homogeneous systems of linear equations: Example 1

Given the homogeneous system of linear equations

$$\left. \begin{array}{rrrrrr} & x_2 & + & x_3 & & = & 0 \\ x_1 & & + & x_3 & & + & x_5 = 0 \\ x_1 & & & & + & x_4 & + & x_5 = 0 \\ & x_2 & & & + & x_4 & & = 0 \end{array} \right\}$$

a routine Gauss elimination yields:

$$\left. \begin{array}{rrrrrr} x_1 & & & + & x_4 & + & x_5 = 0 \\ & x_2 & & + & x_4 & & = 0 \\ & & x_3 & - & x_4 & & = 0 \\ & & & & & 0 & = 0 \end{array} \right\}$$

(Exercise: check this!)

5 Space of solutions of a homogeneous systems of linear equations: Example 1 (continued)

Our system in row echelon form looks like this:

$$\left. \begin{array}{rrrrrr} x_1 & & & + & x_4 & + & x_5 = 0 \\ & x_2 & & + & x_4 & & = 0 \\ & & x_3 & - & x_4 & & = 0 \\ & & & & & 0 & = 0 \end{array} \right\}$$

So we have two parameters, *viz.*, the variables x_4 and x_5 , and the algorithm produces two basis vectors (for the choices $x_4 = 1, x_5 = 0$ and $x_4 = 0, x_5 = 1$, respectively):

$$b_1 = \begin{pmatrix} -1 \\ -1 \\ 1 \\ \textcolor{red}{1} \\ 0 \end{pmatrix} \quad \text{and} \quad b_2 = \begin{pmatrix} -1 \\ 0 \\ 0 \\ \textcolor{red}{0} \\ 1 \end{pmatrix}$$

The space of solutions has basis b_1, b_2 , and is thus a 2-dimensional subspace of $\mathbb{R}^5$.

6 Space of solutions of a homogeneous systems of linear equations: Example 2

Given the homogeneous system of linear equations

$$\left. \begin{array}{ccccccccc} x_1 & + & 2x_2 & + & x_3 & + & x_4 & = & 0 \\ -x_1 & - & 2x_2 & & & + & x_4 & = & 0 \\ -2x_1 & - & 4x_2 & - & 5x_3 & + & x_4 & = & 0 \\ 3x_1 & + & 6x_2 & + & 2x_3 & + & 19x_4 & = & 0 \end{array} \right\}$$

we perform Gauss elimination as usual and obtain:

$$\left. \begin{array}{ccccccccc} x_1 & + & 2x_2 & + & x_3 & + & x_4 & = & 0 \\ & & & & x_3 & + & 2x_4 & = & 0 \\ & & & & & & 9x_4 & = & 0 \\ & & & & & & 0 & = & 0 \end{array} \right\}$$

(Exercise: check this!)

7 Space of solutions of a homogeneous systems of linear equations: Example 2 (continued)

Our system in row echelon form looks like this:

$$\left. \begin{array}{ccccccccc} x_1 & + & 2x_2 & + & x_3 & + & x_4 & = & 0 \\ & & & & x_3 & + & 2x_4 & = & 0 \\ & & & & & & 9x_4 & = & 0 \\ & & & & & & 0 & = & 0 \end{array} \right\}$$

So we have one parameter, *viz.*, the variable x_2 . The algorithm produces a single basis vector (for the choice $x_2 = 1$):

$$b_1 = \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

So the set of solutions has basis b_1 and is thus 1-dimensional.

8 Finding a basis for the set of solutions of a homogeneous systems of linear equations: Justification

Why does the algorithm produce a basis for L ?

First, we may assume (by re-numbering the variables) that the last k variables are the parameters, so that $x_1, x_2, \dots, x_{n-k}$ are *not* parameters. Let b_j be the vector obtained by setting the j th parameter $x_{n-k+j} = 1$, and all other parameters to 0.

These vectors must be linearly independent: look at components $n - k + 1, n - k + 2, \dots, n$, where the b_j "look like unit vectors". (The last k entries of b_{n-k+1} are e_1 , the last k entries of b_{n-k+2} are e_2 and so on. See the example on Slide 5.)

So we need to check that $b_1, b_2, \dots, b_k$ generate L .

9 Finding a basis for the set of solutions of a homogeneous systems of linear equations: Justification (continued)

Let $x \in L$ have components $x_1, x_2, \dots, x_n$. Then x is the solution corresponding to the parameter values $x_{n-k+1}, x_{n-k+2}, \dots, x_n$.

But the linear combination

$$x_{n-k+1}b_1 + x_{n-k+2}b_2 + \dots + x_nb_k$$

is another description of the *same* solution (same parameter values!), so equals x . In other words, we have

$$x \in \text{span} \{b_1, b_2, \dots, b_k\}$$

so that the b_j generate L as required.

10 The dimension of the set of solutions of a homogeneous system of linear equations

We have now proved all the claims from slide 2.

Note that the space L does *not* depend on the details of the Gauss elimination procedure. Hence:

The number of parameters is independent of how exactly the row echelon form is achieved, and is also independent of possible re-ordering of the variables.

To explain the second part of the statement, it is clear that listing the variables in a different order does not change the space L : for example, the equations $x + 2y = 0$ and $2y + x = 0$ have the same solutions (x, y) . Hence $\dim(L)$ does not change under re-ordering. **Which** variables are parameters may be affected, but **how many** variables are parameters does not change!

Section 7.2: The rank of a matrix

11 Column space and rank of a matrix

Let A be an $(m \times n)$ -matrix of real numbers. The columns of A , denoted $c_1, c_2, \dots, c_n$, are vectors in $\mathbb{R}^m$.

The *column space* of A is the span of the columns of A , considered as vectors in $\mathbb{R}^m$:

$$\text{col}(A) = \text{span} \{c_1, c_2, \dots, c_n\} .$$

The *rank* of A , denoted $\text{rank}(A)$, is the dimension of the column space of A :

$$\text{rank}(A) = \dim \text{col}(A)$$

12 Column space and rank of a matrix

Let A be an $(m \times n)$ -matrix of real numbers. The rank of the matrix A satisfies the conditions

$$0 \leq \text{rank}(A) \leq \min(m, n) .$$

Proof. Let $c_1, c_2, \dots, c_n \in \mathbb{R}^m$ be the columns of A as before. By definition $\text{rank}(A) = \dim \text{col}(A) = \dim \text{span} \{c_1, c_2, \dots, c_n\}$, so we want to prove the inequalities

$$0 \leq \dim \text{span} \{c_1, c_2, \dots, c_n\} \leq \min(m, n) .$$

Now $\text{span} \{c_1, c_2, \dots, c_n\}$ is a subspace of $\mathbb{R}^m$, hence its dimension cannot exceed m (slide 28 of Chapter 6). On the other hand, the set $\{c_1, c_2, \dots, c_n\}$ contains a basis for its span (by basis selection algorithm), so its dimension cannot exceed n either. This proves the second inequality. — The first inequality holds by the definition of dimension. ■

Section 7.3: The Kronecker-Capelli theorem

13 When does a system have a solution?

Given the system of linear equations

$$\left. \begin{array}{cccccc} a_{1,1}x_1 & + & a_{1,2}x_2 & + & \dots & + & a_{1,n}x_n & = & b_1 \\ a_{2,1}x_1 & + & a_{2,2}x_2 & + & \dots & + & a_{2,n}x_n & = & b_2 \\ \vdots & & \vdots & & \ddots & & \vdots & = & \vdots \\ a_{m,1}x_1 & + & a_{m,2}x_2 & + & \dots & + & a_{m,n}x_n & = & b_m \end{array} \right\} \quad (\text{S})$$

let A and $\tilde{A}$ denote its matrix of coefficients and its augmented matrix of coefficients. That is,

$$A = \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{pmatrix} \quad \tilde{A} = \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} & b_1 \\ \vdots & \ddots & \vdots & \vdots \\ a_{m,1} & \cdots & a_{m,n} & b_m \end{pmatrix}$$

14 When does a system have a solution? (ct'd)

We let $c_1, c_2, \dots, c_n, b$ denote the columns of $\tilde{A}$, that is,

$$c_1 = \begin{pmatrix} a_{1,1} \\ a_{2,1} \\ \vdots \\ a_{m,1} \end{pmatrix}, \quad c_2 = \begin{pmatrix} a_{1,2} \\ a_{2,2} \\ \vdots \\ a_{m,2} \end{pmatrix}, \quad \dots, \quad c_n = \begin{pmatrix} a_{1,n} \\ a_{2,n} \\ \vdots \\ a_{m,n} \end{pmatrix}, \quad b = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

With this notation, the following two statements are equivalent (just a matter of using different notation):

- (1) The system (S) has a solution
- (2) $b \in \text{span} \{c_1, c_2, \dots, c_n\}$ ($= \text{col}(A)$)

Now $\text{col}(\tilde{A}) = \text{span} \{c_1, c_2, \dots, c_n, b\}$ contains $\text{col}(A)$, so statement (2) is equivalent to

- (3) $\text{col}(\tilde{A}) = \text{col}(A)$

which by slide 28 of Chapter 6 says the same as

- (4) $\dim \text{col}(A) = \dim \text{col}(\tilde{A})$.

15 When does a system have a solution? (ct'd)

Recalling the definition of "rank" of a matrix, this shows part 1 of the following result (Kronecker-Capelli theorem):

- The system (S) on slide 13 has a solution if and only if its matrix of coefficients A and its augmented matrix of coefficients $\tilde{A}$ have the same rank.
- Let L denote the space of solutions of the associated *homogeneous* system (with matrix of coefficients A). Then

$$\dim L + \text{rank}(A) = n .$$

16 When does a system have a solution? (ct'd)

Remark: We know that $\dim(L)$ is precisely the number of parameters (determined by bringing the system into row echelon form), see slide 2 of this chapter. Hence the number of non-parameters is $n - \dim(L)$, and the latter is the rank of A by second claim of previous slide.

$\text{rank}(A)$ is the number of non-parameters, as determined by bringing A into row echelon form. In particular, the number of non-parameters is independent of how Gauss elimination is performed, and does not change when re-ordering the variables.

17 Sketch argument for claim 2 on slide 15

- The columns $c_1, \dots, c_n$ of A generate the subspace $\text{col}(A)$.
- Any linear dependence relation $x_1c_1 + x_2c_2 + \dots + x_nc_n = \underline{0}$ corresponds precisely to a solution of the *homogeneous* system with matrix of coefficients A .
- We do not change the solutions when bringing the matrix A into row echelon form B .
- Thus the linear dependence relations of the columns of A are *exactly the same* as the linear dependence relations for the columns of B .

- In B , the columns which incorporate a non-parameter in first position form a basis of $\text{col}(B)$.
- Hence the *corresponding* columns of A form a basis for $\text{col}(A)$. Thus $\text{rank}(A) = \dim \text{col}(A)$ is precisely the number of non-parameters.
- $\dim(L) + \text{rank}(A) = n$ follows as n is the sum of the number of parameters and the number of non-parameters.

18 Explicit example

We want to study the following system of linear equations:

$$\left. \begin{array}{rrrr} 2x_1 & + & x_2 & + & x_3 & = & 1 \\ 2x_1 & + & x_2 & + & 4x_3 & = & 3 \\ -4x_1 & - & 2x_2 & - & 5x_3 & = & -4 \end{array} \right\} \quad (\text{S})$$

Bringing this into row echelon form results in the following system:

$$\left. \begin{array}{rrrr} 2x_1 & + & x_2 & + & x_3 & = & 1 \\ & & & & 3x_3 & = & 2 \\ & & & & 0 & = & 0 \end{array} \right\}$$

Thus (S) has a solution. Hence by part 1 of Theorem, the matrices

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 2 & 1 & 4 \\ -4 & -2 & -5 \end{pmatrix} \quad \text{and} \quad \tilde{A} = \begin{pmatrix} 2 & 1 & 1 & 1 \\ 2 & 1 & 4 & 3 \\ -4 & -2 & -5 & -4 \end{pmatrix}$$

have the same rank.

19 Explicit example (ct'd)

There is one parameter (*viz.*, x_2), so if L denotes the space of solutions of the associated homogeneous system

$$\left. \begin{array}{rrrr} 2x_1 & + & x_2 & + & x_3 & = & 0 \\ 2x_1 & + & x_2 & + & 4x_3 & = & 0 \\ -4x_1 & - & 2x_2 & - & 5x_3 & = & 0 \end{array} \right\} \quad (\text{H})$$

then we know $\dim L = 1$.

By part 2 of the theorem, the rank of A is thus

$$3 - \dim L = 2 ,$$

which is indeed the number of non-parameters.

20 Explicit example (ct'd)

In the row echelon form, the non-parameters sit in columns 1 and 3:

$$\left. \begin{array}{rclcl} 2x_1 & + & x_2 & + & x_3 & = & 1 \\ & & & & 3x_3 & = & 2 \\ & & & & 0 & = & 0 \end{array} \right\}$$

Thus the 1st and 3rd column of A are a basis for the column space $\text{col}(A)$ of A :

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 2 & 1 & 4 \\ -4 & -2 & -5 \end{pmatrix} \quad \text{col}(A) = \text{span} \left\{ \begin{pmatrix} 2 \\ 2 \\ -4 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \\ -5 \end{pmatrix} \right\}$$

Of course the same two vectors are a basis for $\text{col}(\tilde{A})$ in this case, as $\text{col}(A) = \text{col}(\tilde{A})$.

Section 7.1: A simplified basis selection algorithm in $\mathbb{R}^m$

21 Basis selection for subspaces of $\mathbb{R}^m$

As an application of the arguments from slide 17, we obtain a much simplified basis selection algorithm for subspaces of $\mathbb{R}^m$:

Let $c_1, c_2, \dots, c_n$ be elements of $\mathbb{R}^m$. We can determine a basis for the vector space

$$U = \text{span} \{c_1, c_2, \dots, c_n\}$$

as follows:

- Define A to be the m -by- n -matrix with j th column c_j , so $U = \text{col}(A)$.
- Use Gauss elimination to transform A into a matrix B which is in row echelon form.
- Call c_j a *nice vector* if in some row of B , the first non-zero entry is in column j .
- The set of nice vectors is a basis for U .

22 Example for basis selection in $\mathbb{R}^4$

We want to determine a basis for the subspace U of $\mathbb{R}^4$ spanned by the following six vectors:

$$c_1 = \begin{pmatrix} 4 \\ 4 \\ 2 \\ 6 \end{pmatrix} \quad c_2 = \begin{pmatrix} 2 \\ 3 \\ 0 \\ 3 \end{pmatrix} \quad c_3 = \begin{pmatrix} 2 \\ 1 \\ 2 \\ 3 \end{pmatrix}$$

$$c_4 = \begin{pmatrix} 1 \\ 1 \\ 11 \\ 1 \end{pmatrix} \quad c_5 = \begin{pmatrix} 3 \\ 1 \\ -7 \\ 5 \end{pmatrix} \quad c_6 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

We define a 4×6 -matrix A with these vectors as columns:

$$A = \begin{pmatrix} 4 & 2 & 2 & 1 & 3 & 0 \\ 4 & 3 & 1 & 1 & 1 & 1 \\ 2 & 0 & 2 & 11 & -7 & 1 \\ 6 & 3 & 3 & 1 & 5 & 1 \end{pmatrix}$$

23 Example for basis selection in $\mathbb{R}^4$ (ct'd)

The Gauss algorithm produces the following matrix from A :

$$B = \begin{pmatrix} 4 & 2 & 2 & 1 & 3 & 0 \\ 0 & 1 & -1 & 0 & -2 & 1 \\ 0 & 0 & 0 & \frac{21}{2} & -\frac{21}{2} & 2 \\ 0 & 0 & 0 & 0 & 0 & \frac{23}{21} \end{pmatrix}$$

The first non-zero entries (marked in green) occur in columns 1, 2, 4 and 6. Thus the vectors c_1 , c_2 , c_4 and c_6 are marked "nice".

Result: $U = \text{span}\{c_1, c_2, \dots, c_6\}$ has basis consisting of the four vectors

$$c_1 = \begin{pmatrix} 4 \\ 4 \\ 2 \\ 6 \end{pmatrix} \quad c_2 = \begin{pmatrix} 2 \\ 3 \\ 0 \\ 3 \end{pmatrix} \quad c_4 = \begin{pmatrix} 1 \\ 1 \\ 11 \\ 1 \end{pmatrix} \quad c_6 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$$

Since $\dim(U) = 4 = \dim(\mathbb{R}^4)$ we also see that $U = \mathbb{R}^4$.

24 Summary

- The space of solutions of a homogeneous system of linear equations is finitely generated. Its dimension is precisely the number of parameters of the system.
- Every subspace U of a finitely generated vector space V is finitely generated; moreover, $\dim(U) \leq \dim(V)$, with equality if and only if $U = V$.
- We can use the theory of dimension to formulate a criterion for when a system of linear equations has a solution, in terms of the rank of matrices (Kronecker-Capelli theorem).
- We can use Gauss elimination to get a convenient version of the Basis Selection Algorithm for the vector space $\mathbb{R}^m$.